MIT SCHOOL OF COMPUTING
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MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: Programming Language II Roll No.: 60 Exp. No.: –
Name of the Experiment: Assignment - 02
Performed on:
Submitted on:

Marks	Teacher's Signature with date
8 / 10	1/12/24

1. List the python sequence data type and state one key characteristics of each with respect to mutability?

Ans→ (i) List:- Ordered and mutable (elements can be changed)
(ii) Tuple:- Ordered and immutable (cannot be modified)
(iii) string:- Ordered and immutable sequence of characters
(iv) Set:- Unordered, mutable, store unique elements only.
(v) Dictionary:- ordered (Python 3.7+), mutable, stores data as key value pairs.

2. Explain how indexing and silicing work in python sequences. Illustrate how they differ when applied to string and list?

Ans→ Indexing:- It is used to access a single element using its position (start from 0)
Slicing:- It is used to access a part of the sequence using [start : end : step]

Ex→ S = "Python"
I = [10, 20, 30, 40]

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

3. Write a python program that:
a) Uses of indexing and slicing on a string list, and.
b) Applies at least two built in function to apply display meaningful results.

Ans → text = "Vidhan"
print ("first character:", text[0])
print ("Substring:", text[0:3]
print ("length of string:", len(text))
print ("Uppercase:", text upper())

Output:
First character : V
Substring: Vid
length of string : 7
Uppercase : VIDHAN

4. Analyze the difference between equality (==) and identity (is) when used with sequence type. Discuss how object mutability and memory refrence affect program behaviour?

Ans → Equality (==) checks whether values are same
Identity (is) checks whether both variable print to the same memory location.

Ex→
a=[1,2,3]
b=[1,2,3]
print (a==b)
print (a is b)
Output:
True
False

Explanation→
- Lists are mutable and stored in different memory location
- (==) compares content
- (is) compares memory reference.

5. Evaluate the suitability of lists, sets, and dictionaries for storing and managing student records. Justify which structure is most efficient and why?

Ans → List:- Allows duplicates, slower searching
Set:- Stores only unique value, no indexing.
Dictionaries:- stores unique student IDs as keys and detail as value
Most efficient:- Dictionary, because it allow fast access, unique keys and structured data storage.

6. Design and implement a python program that uses:
- Appropriate sequence type
- Indexing or slicing
- Built in function

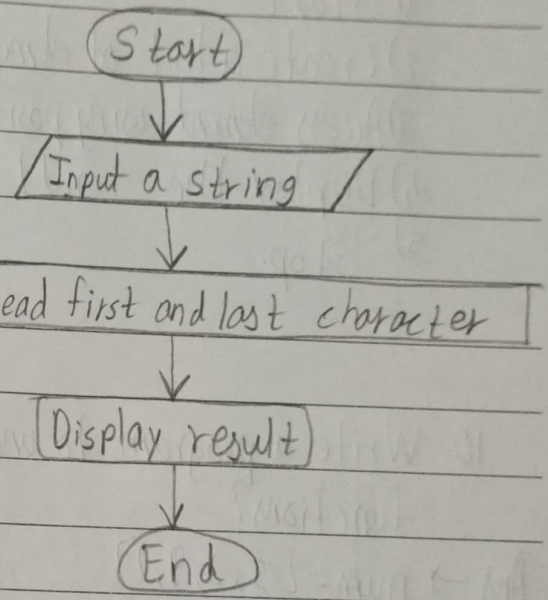
to manage and process student data efficiently.

Ans →
```
Students = {101: 'Amit', 102: "Neha", 103: "Rahul"}
print ("Student 101:", students [101])
print ("Total students:", len (students))
print ("Student IDs:", list (students.key ()))
```
Output:-
Student 101: Amit
Total students: 3
Student IDs: [101, 102, 103]

7. Write a program to point the first and last character of a string?

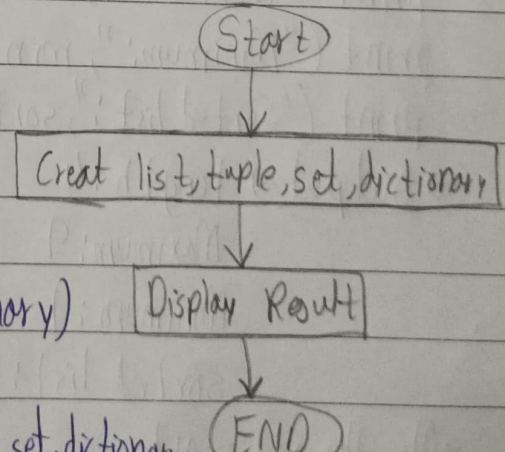
Ans→ S= input ("Enter a string:")
print ("first character:",s[0])
print ("last character:",s[-1])
→ output: First character: p
Last character: n

→ Algorithm:-
1) Start
2) Read a string
3) Find the first character
4) Find the last character
5) Display result.
6) Stop



8) Write a program to create a list, tuple, set and dictionary?

Ans→ List=[1,2,3]
tuple = (4,5,6)
String = [7,8,9]
dictionary = {1:"One", 2:"Two"}
print (list, tuple, string, dictionary)



→ Algorithm:- 1) Start
2) Create a list, tuple, set, dictionary
3) Display result
4) Stop

Output → [1,2,3]
(4,5,6)